Intermittent Fasting Enhances Leptin Sensitivity via JMJD3-Mediated Epigenetic Regulation

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INTRODUCTION

Leptin, a hormone produced by adipose tissue that plays a vital role in maintaining energy balance by suppressing appetite and increasing energy expenditure. Under normal physiological conditions, leptin signals to the hypothalamus to suppress food intake and body weight reductions. However, in obesity, a state of leptin resistance often arises where the hypothalamus becomes unresponsive to leptin signaling impairing this regulatory mechanism and contributing to further weight gain and metabolic dysfunction. Intermittent fasting (IF), a dietary intervention involving alternating periods of feeding and fasting, has gained attention for its metabolic benefits, including weight control and improved insulin sensitivity. Despite growing interest, the effects of IF on leptin sensitivity, especially the molecular mechanisms involved, remain poorly understood. This research investigates whether IF enhances leptin responsiveness by regulating the leptin receptor (LepRb) expression in the hypothalamus. Specifically, we focused on the role of H3K27me3, a repressive epigenetic mark known to suppress gene transcription. We hypothesized that IF reduces H3K27me3 levels at the LepRb promoter, thereby increasing LepRb expression and improving leptin signaling. These findings offer insight into how intermittent fasting may restore leptin sensitivity through epigenetic regulation of hypothalamic leptin receptor expression and highlight potential targets for future pharmacological and nutritional strategies to combat obesity and metabolic disorders.

OBJECTIVES

1. To investigate the molecular mechanism behind body weight management through IF.
2. To evaluate the impact of IF on hypothalamic LepRb expression and downstream leptin signaling.
3. To examine the role of epigenetic modifications, particularly H3K27me3, in regulating LepRb transcription and signaling.
4. To develop nutritional and pharmacological strategies that enhance leptin sensitivity and prevent early obesity and type 2 diabetes.

MATERIALS AND METHODS

Animals. 7-week-old male C57BL/6J mice were obtained and maintained on a high-fat diet (HFD; 60% kcal from fat; Research Diets, New Brunswick, NJ) for 10 weeks to induce diet-induced obesity. Following this period, mice were randomly assigned to either an ad libitum (AL; free access to food) group or an intermittent fasting (IF) group, which followed an alternate-day fasting regimen (24 h fasting followed by 24 h feeding) for 20 days (10 cycles). All animal procedures complied with institutional and national ethical guidelines and were approved by the Institutional Animal Care and Use Committee (IACUC) at Arizona State University (Protocol #25-2125R).

Leptin administration. Mice were intraperitoneally injected with recombinant leptin (0.25 mg/kg body weight) at 6:00 PM and 12:00 AM daily for 3 days. Body weight and food intake were recorded throughout the treatment period.

Chromatin immunoprecipitation (ChIP)-PCR: Hypothalamic tissues were treated with 1% formaldehyde for 10 min. Genomic DNA was extracted and sheared into 200-500 bp fragments using a Q800R sonicator (QSONICA). Chromatin was immunoprecipitated using specific antibodies, and cross-links were reversed by incubation at 65° C for 4 hours. DNA was purified using commercial kits or standard phenol-chloroform extraction and subjected to qPCR analysis using *LepR* 5′ and 3′ primers.

Stereotaxic microinjection and AAV transduction. Mice were anesthetized and microinjected bilaterally in the mediobasal hypothalamus with AAV vectors carrying either GFP-U6-Jmjd3 shRNA or GFP-U6-scramble shRNA (control). Mice were allowed to recover fully before undergoing experimental procedures.

RESULTS

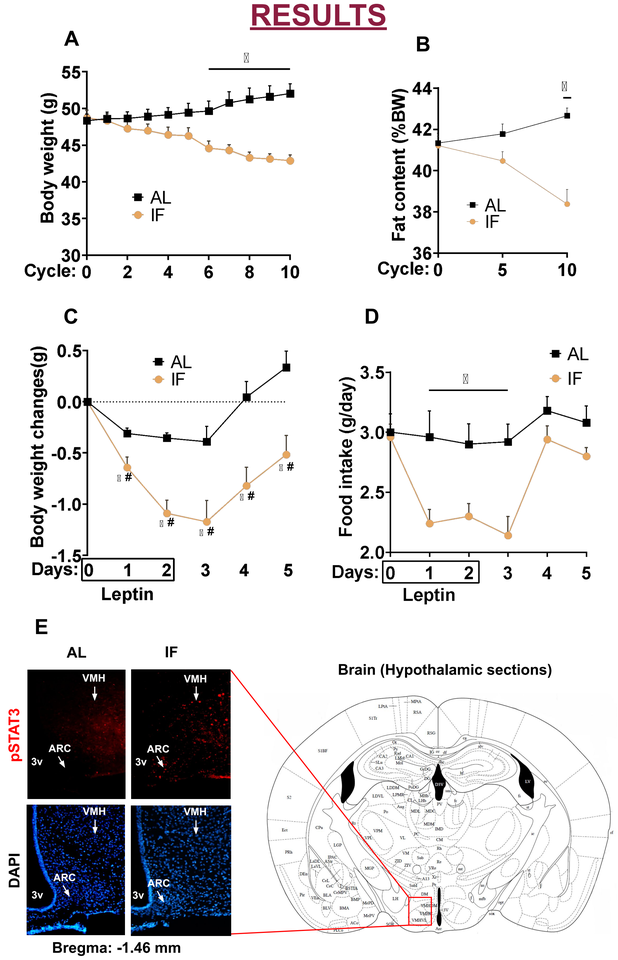


Figure 1. IF increases hypothalamic LepRb expression independent of body weight and fat content. (A-E) Male C57BL/6J mice were fed on a HFD for 10 weeks (n=8 per group) and then randomly assigned to either AL or IF group for 20 days (10 cycles). (**A**) Body weight. (**B**) Fat content measured by EchoMRI. (**C-D**) Mice were i.p. injected with leptin (0.25 mg/kg BW) for 3 days. Body weight changes and food intake levels were measured. (**E**) Hypothalamic sections (Bregma –1.46 mm) were immunostained with anti-pSTAT3 antibody to detect pSTAT3⁺ signals (red), and nuclei were counterstained with DAPI (blue). Arcuate nucleus (ARC) and ventromedial hypothalamus (VMH) areas are shown. Values are presented as mean ± SEM. *p < 0.05

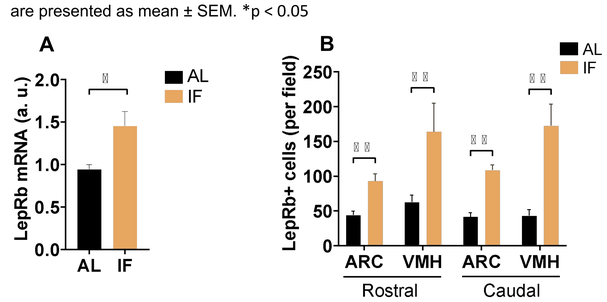


Figure 2. IF enhances hypothalamic LepRb expression. Male C57BL/6J mice were fed on a HFD for 10 weeks (n=8 per group) and then randomly assigned to either AL or IF group for 10 days (5 cycles). (**A**) *LepRb* mRNA expression measured by qPCR. (**B**) Quantification of LepRb+ cells in the Arcuate nucleus (ARC) and ventromedial hypothalamus (VMH) at both rostral and caudal levels. Values are presented as mean ± SEM. *p < 0.05 **p < 0.01

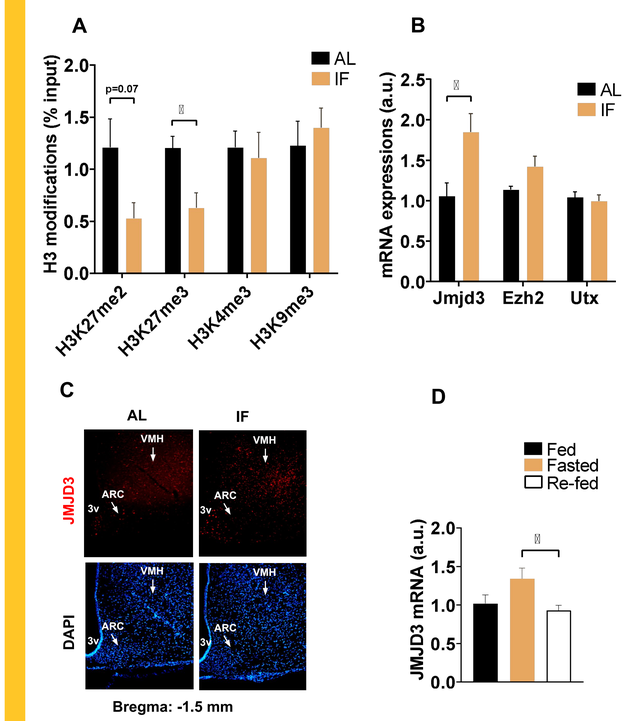
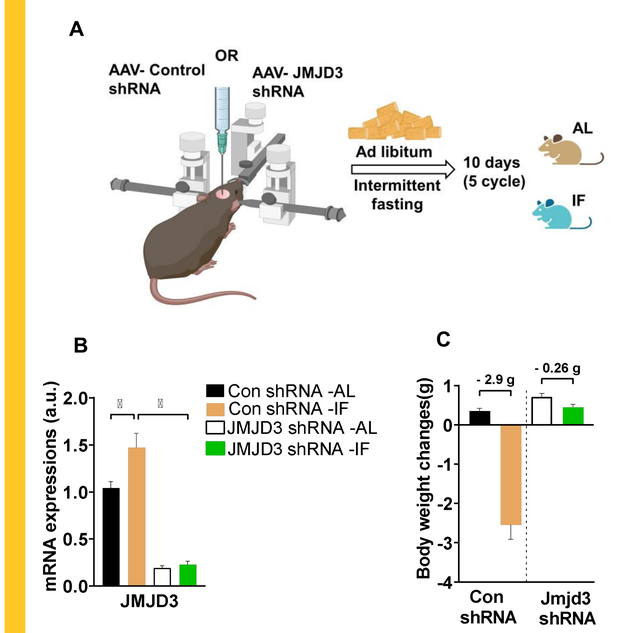


Figure 3. IF modulates Jmjd3 expression and hypothalamic H3K27m3 modification. (A-C) Male C57BL/6J mice were fed on a HFD for 10 weeks (n=4 to 8 per group) and then randomly assigned to either AL or IF group for 20 days (10 cycles). (**A**) H3K27 methylation levels of the LepRb promoter measured by ChIP-PCR. (**B**) Hypothalamic gene expressions of histone demethylases were measured by qPCR. © Hypothalamic sections (Bregma -1.5 mm) were immunostained with anti-JMJD3 antibody, and JMJD3⁺ signals (red) were quantified. Nuclei were counterstained with DAPI (blue). Arcuate nucleus (ARC) and ventromedial hypothalamus (VMH) areas are shown. (**D**) JMJD3 mRNA levels in hypothalamic tissue under fed, fasted (24 hours), and followed by re-fed (6 hours) conditions. Values are presented as mean ± SEM. *p < 0.05



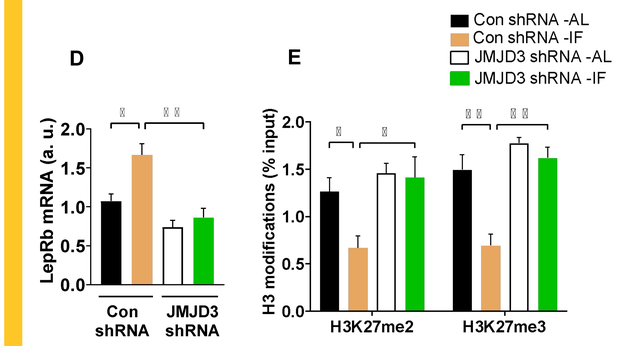


Figure 4. JMJD3 knockdown mediates IF-induced upregulation of LepRb and histone demethylation. (A–E) Male C57BL/6J mice were bilaterally microinjected in the mediobasal hypothalamus with AAV vectors carrying either control shRNA or JMJD3 shRNA. After recovery, mice were fed a high-fat diet (HFD) for 10 weeks (n = 5 per group) and then randomly assigned to either an AL or IF group for 10 days (5 cycles). **(A)** Schematic overview of experimental design and AAV delivery strategy. **(B)** Hypothalamic JMJD3 mRNA expression in control and JMJD3 shRNA groups was measured by qPCR. © Body weight changes were measured after 5 cycles of intermittent fasting (IF) in control and JMJD3 knockdown mice. **(D)** *LepRb* mRNA expression measured by qPCR. **(E)** H3K27me2 and H3K27me3 enrichment at the LepRb promoter measured by ChIP-qPCR in control and JMJD3 knockdown mice. Values are presented as mean ± SEM. *p < 0.05 **p < 0.01

CONCLUSIONS

1. Intermittent fasting (IF) improves leptin sensitivity in obese mice without altering overall body weight or fat mass.
2. IF enhances hypothalamic LepRb expression and leptin signaling, evidenced by increased STAT3 phosphorylation and target gene activation.
3. Epigenetic analysis revealed reduced H3K27 methylation at the LepRb promoter, suggesting that IF promotes transcriptional activation.
4. These results support IF as an effective dietary approach to reverse leptin resistance and restore metabolic balance in obesity.

REFERENCES

1. Gao W, Liu JL, Lu X, Yang Q. Epigenetic regulation of energy metabolism in obesity. *Journal of Molecular Cell Biology*. 2021;13(7):480-499. doi:10.1093/jmcb/mjab043
2. Endo S, Uto A, Miyashita K, et al. Intermittent fasting sustainably improves glucose tolerance in normal weight male mice through Histone hyperacetylation. *Journal of the Endocrine Society*. 2023;7(7). doi:10.1210/jendso/bvad082

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